Problem Set #3 - Answer Sheet

1.

See the code file

2.
a.

Expection: $E[X] = np = 18 \times 0.16 = 2.88$

Mode: $M[X] = 3$, where $M[X]$ is the mode of X

Probability of Mode: $P(X = 3) = \binom{18}{3} \times (0.16)^3 \times (0.84)^{15} = 0.2423$

Probability of two other integers:

$$P(X = 2) = \binom{18}{2} \times (0.16)^2 \times (0.84)^{16} = 0.2383$$

$$P(X = 4) = \binom{18}{4} \times (0.16)^4 \times (0.84)^{14} = 0.173$$

b.

Expection: $E[X] = \frac{1}{p} = 5$

Mode: $M[X] = 1$

Probability of Mode: $P(X = 1) = (1 - p)^{1-1}p = 0.2$

Probability of two other integers:

$$P(X = 0) = 0$$

$$P(X = 5) = (1 - p)^{5-1}p = 0.08192$$

c.

Expection: $E[X] = \frac{r}{p} = \frac{300}{19}$

Mode: $M[X] = 8$

Probability of Mode: $P(X = 8) = \binom{8-1}{3-1} p^3(1-p)^{8-3} = 0.0502$

Probability of two other integers:

$P(X = 7) = \binom{7-1}{3-1} p^3(1-p)^{7-3} = 0.0443$

$P(X = 15) = \binom{15-1}{3-1} p^3(1-p)^{15-3} = 0.0345$

3.

a. $S\$4.19539$

Let

M: A random variable represents the money can the driver make on this trip

X: A random variable represents the number of users will request in the next five minutes

Based on the information given in the problem, we obtain:

$X \sim Poi(2)$

Therefore

$E[M] = P(X = 0) \times (6 - 7) + P(X = 1) \times (6 \times 2 - 7) + P(X = 2) \times (6 \times 3 - 7) = 4.19539$

b. 0.1813

Let

T: A random variable represents seconds until another passenger makes a request

Based on the information given in the problem, we obtain:

$T \sim Exp(\frac{2}{5 \times 60})$

Therefore

$$F(t=30)=1-e^{-\frac{1}{150}\times 30}=0.1813$$

4. 0.9118

Let

X: A random variable represents the number of judges who votes

Based on the information given in the problem, we obtain:

$$X\sim Bin(12,0.85)$$

Therefore

$$P(9\leq X\leq 12)=1-F(8)=0.9118$$

5. 2.589

Let

N: A random variable represents the number of tests necessary for each group of 6 people

X: A random variable represents the number of people having measle in the group of 6 people

Based on the information given in the problem, we obtain:

$$X\sim Bin(6,0.05)$$

Therefore

$$E[N]=P(X=0)\times 1+(1-P(X=0))\times(1+6)=2.589$$

6.

a. $\frac{3}{8}$

Because

$$\int_{-1}^{1} c(2-2x^2)dx=1$$

Therefore

$$c = \frac{3}{8}$$

b.

When $-1 < x < 1$, $F(x) = \int_{-1}^{x} c(2 - 2x^2)dx = -\frac{1}{4}x^3 + \frac{3}{4}x + \frac{1}{2}$

Therefore

$$F(x) = \begin{cases} 0, & x \leq -1 \\ -\frac{1}{4}x^3 + \frac{3}{4}x + \frac{1}{2}, & -1 < x < 1 \\ 0, & x \geq 1 \end{cases}$$

c. 0

$$E[X] = \int_{-1}^{1} xf(x)dx = 0$$

7.
a.

Let

T: A random variable represents the hours until the bug manifests

Based on the information given in the problem, we obtain:

$$T \sim Exp(\frac{2}{400})$$

Therefore

$$F(3) = 1 - e^{-0.005 \times 3} = 0.0149$$

b.

Let

X: A random variable represents the number of users experience the bug

$$X \sim Bin(100000, 0.0149)$$

We can further compute that

$$P(x > 10000) = 1 - P(x \leq 10000) = 1 - F(10000)$$

To use a normal approximation to estimate the probability, we first need to calculate the expectation μ and variance σ

$$\mu = np = 1490$$

$$\sigma = np(1-p) = 1467.799$$

Therefore

$$F(10000) = \Phi(\frac{10000 - 1490}{1467.799}) = 0.99999$$

8.

a. $\frac{a+b}{2}$

$$F(x) = 0.5 \Rightarrow x = \frac{a+b}{2}$$

b. $x = \mu$

$$F(x) = \Phi(\frac{x-\mu}{\sigma}) = 0.5 \Rightarrow x = \mu$$

c. $\frac{log2}{\lambda}$

$$F(x) = 1 - e^{-\lambda x} = 0.5 \Rightarrow x = \frac{log2}{\lambda}$$

9.

a. e^{-5}

Let

X: A random variable represents the number of strings hashed into the first bucket

Based on the information given in the problem, we obtain:

$$X \sim Bin(10000, \frac{1}{2000})$$

We can further compute that

$$\lambda = E[X] = np = 5$$

Using the Poisson approximation, we can write

$$X \sim Poi(5)$$

Therefore

$$P(x = 0) = \frac{5^0 \times e^{-5}}{0!} = e^{-5}$$

b. 0.9326

Based on the results of question a, we can further compute that

$$F(x = 8) = \sum_{i=0}^{8} \frac{5^i e^{-5}}{i!} = 0.9326$$

10.

a. 0.8949

Let

X: A random variable represents the number of strings hashed into the first bucket by hash function 1

Y: A random variable represents the number of strings hashed into the first bucket by hash function 2

Z: A random variable represents the number of strings hashed into the first bucket by hash function 3

Based on the information given in the problem, we obtain:

$$X \sim Bin(1000, \frac{1}{9000})$$

$$Y \sim Bin(1000, \frac{1}{9000})$$

$$Z \sim Bin(1000, \frac{1}{9000})$$

Therefore the probability that the first bouket has 0 hashed to it is:

$$p = P(X = 0)P(Y = 0)P(Z = 0) = 0.8949$$

b. 0.000119

Assume this is the $(n+1)$th string hashed to the bucket, then we can turn the question into what's the probability that the 3 buckets correspond to the $(n+1)$th string has been hashed to in the first n hashings.

Based on the result of question a, we can compute the probability through the CDF of X, Y, and Z. However, since the number is large ($n = 1000$), and the value of the probability ($p = \frac{1}{9000}$), we can use Poisson approximation to compute the probability.

We can approximate that

$$X \sim Bin(\frac{1}{9})$$

$$Y \sim Bin(\frac{1}{9})$$

$$Z \sim Bin(\frac{1}{9})$$

Then we can further compute that

$$P(X = 0) = P(Y = 0) = P(Z = 0) = e^{-\frac{1}{9}}$$

Bacause the $(n+1)$th string can be misidentified only when all the buckets have been hashed before, so the probability of misidentification is

$$p = (1 - P(X = 0))(1 - P(Y = 0))(1 - P(Z = 0)) = 0.000119$$

c. 0.106

If there's only one hash function, then the probability of misidentification is

$$p = 1 - P(X = 0) = 0.106$$

which is nearly 900 times larger then the probability of misidentification of 3 hash functions

11.
a. 0.94

Let

CS: A random variable represents the value of Climate Sensitivity

Based on the information given in the problem, we can compute:

$$P(CS \leq 7) = \sum_{i=0}^{7} P(CS = i) = 0.94$$

b. 0.0433**;** 7.2

$$f_1\text{: } \int_{7.5}^{30} \frac{K}{x} dx = 0.06 \Rightarrow K = 0.0433$$

$$f_2\text{: } \int_{7.5}^{30} \frac{K}{x^3} dx = 0.06 \Rightarrow K = 7.2$$

c. 0.0476**;** 0.032

$$f_1\text{: } \int_{10}^{30} \frac{0.0433}{x} dx = 0.0476$$

$$f_2\text{: } \int_{10}^{30} \frac{7.2}{x^3} dx = 0.032$$

d. 3.99425**;** 3.74

f_1:

$$E[X] = (0 \times 0.00 + 1 \times 0.11 + 2 \times 0.26 + 3 \times 0.22 + 4 \times 0.16 + 5 \times 0.09 + 6 \times 0.06 + 7 \times 0.04) + \int_{7.5}^{30} x \frac{0.0433}{x} dx = 3.02 + 0.97425 = 3.99425$$

f_2:

$$E[X] = 3.02 + \int_{7.5}^{30} x \frac{7.2}{x^3} dx = 3.02 + 0.72 = 3.74$$

e.

f_1:

$$E[X^2] = (0^2 \times 0.00 + 1^2 \times 0.11 + 2^2 \times 0.26 + 3^2 \times 0.22 + 4^2 \times 0.16 + 5^2 \times 0.09 + 6^2 \times 0.06 + 7^2 \times 0.04) + \int_{7.5}^{30} x^2 \frac{0.0433}{x} dx = 12.06 + 18.27 = 30.33$$

$$f_2\text{: } E[X^2] = 12.06 + \int_{7.5}^{30} x^2 \frac{7.2}{x^3} dx = 12.06 + 9.983 = 22.043$$